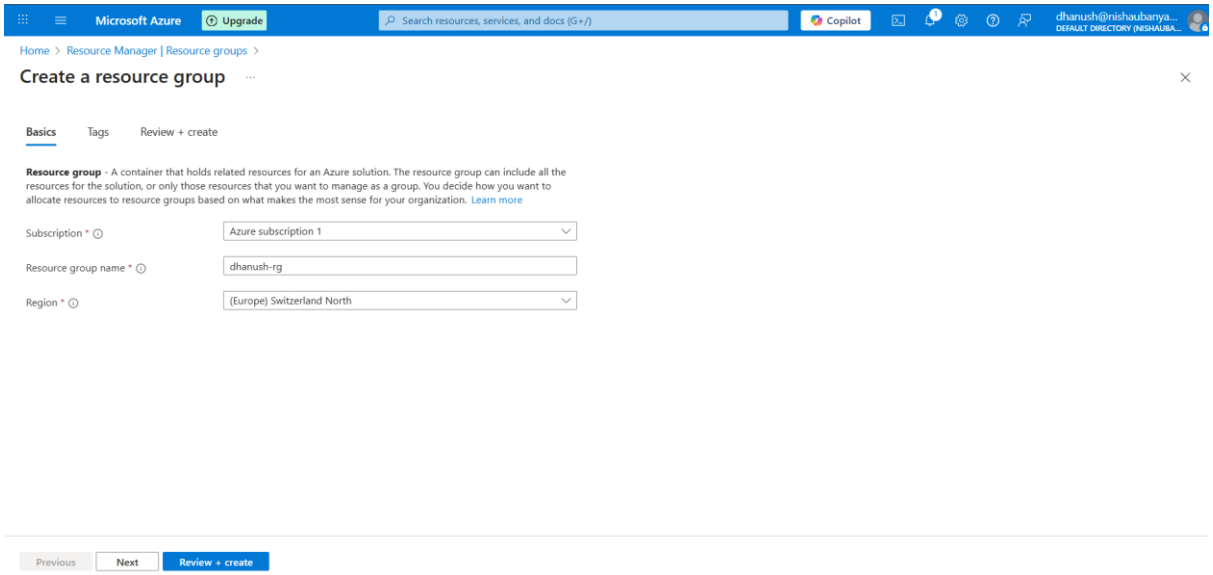
LAB -14 : Create VMSS with Autoscaling

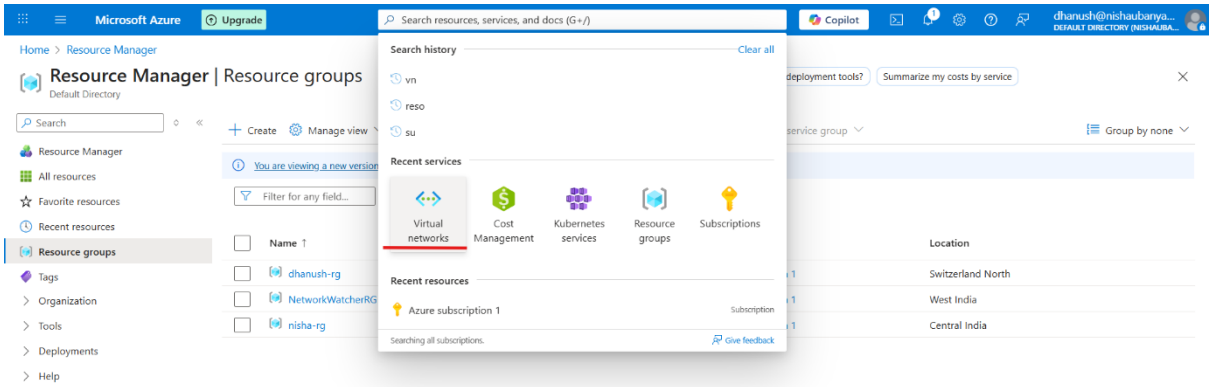
Step 1: Create a Resource Group

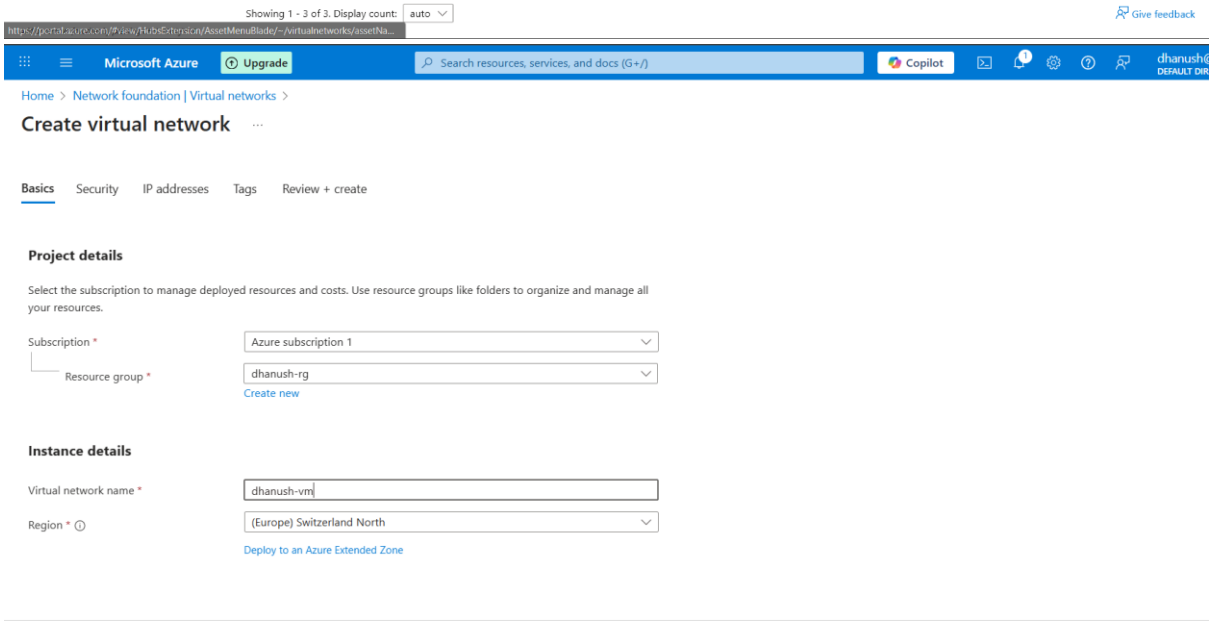
Azure Portal

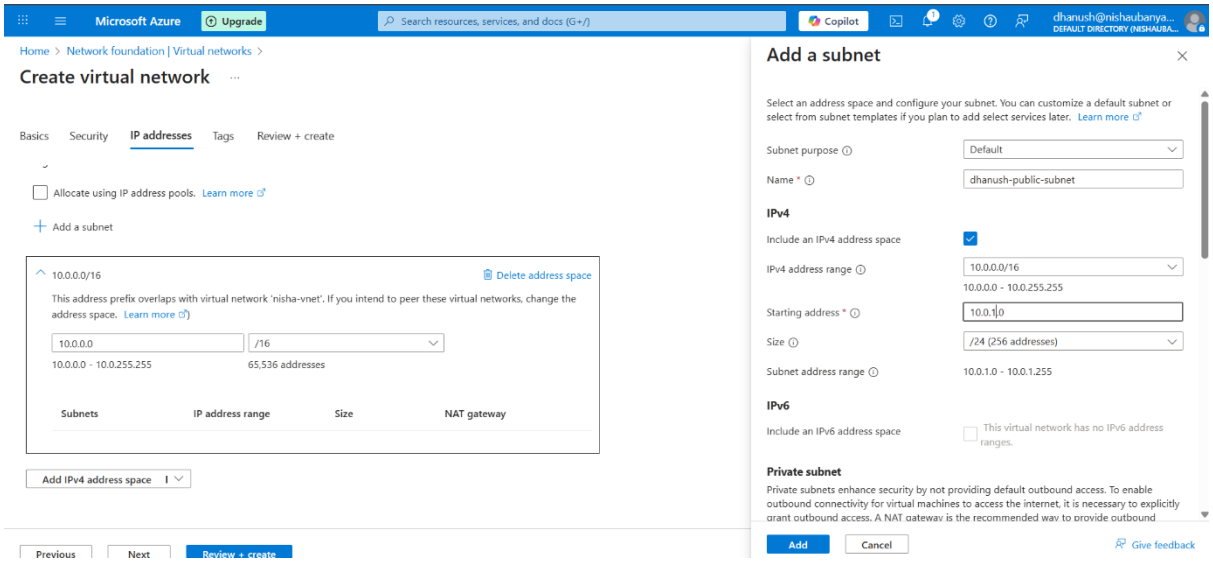
1. Go to **Resource groups → + Create**.
2. Name: vmss-lab-rg
3. Region: Choose nearest region (e.g., East US).
4. Review + Create.

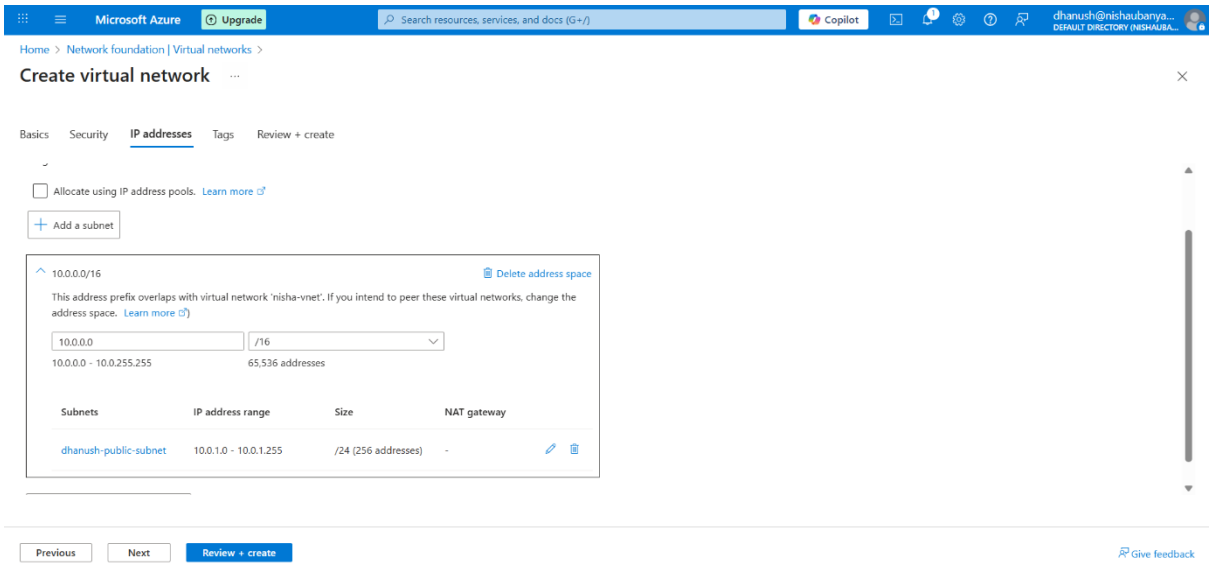


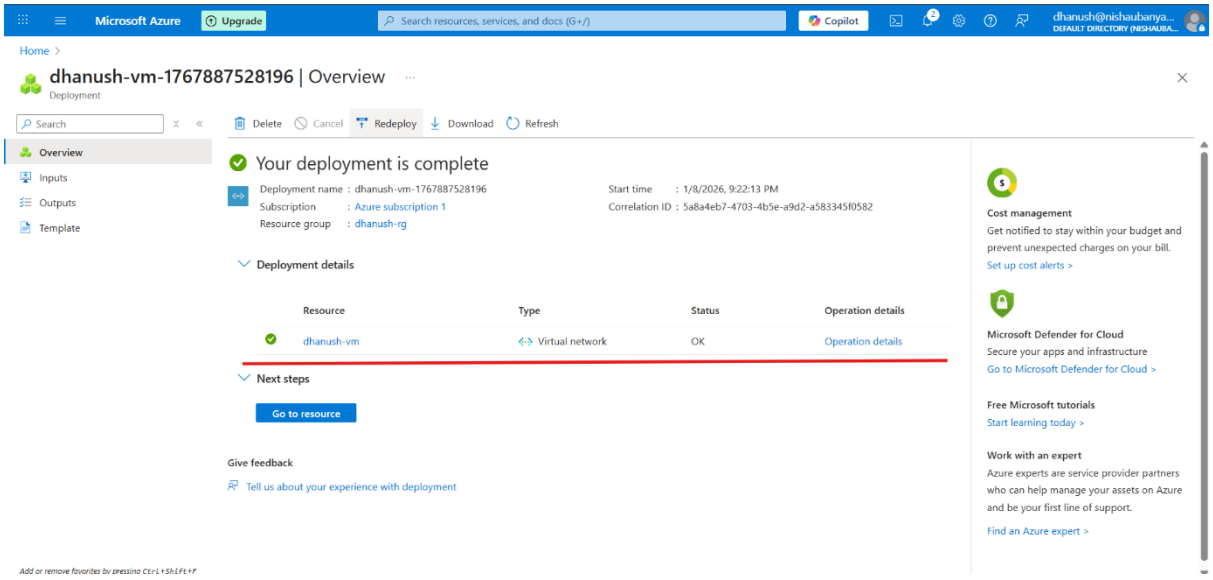
STEP 2 : Create a vnet and subnet


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Resource Manager | Resource groups
Default Directory
Search
Resource Manager
All resources
Favorite resources
Recent resources
Resource groups
Tags
Organization
Tools
Deployments
Help
Create
Manage view
Group by none
Filter for any field...
Name
Location
dhanush-rg
Switzerland North
NetworkWatcherRG
West India
nisha-rg
Central India
Search history
Clear all
vn
reso
su
Recent services
Virtual networks
Cost Management
Kubernetes services
Resource groups
Subscriptions
Recent resources
Azure subscription 1
Subscription
Searching all subscriptions.
Give feedback

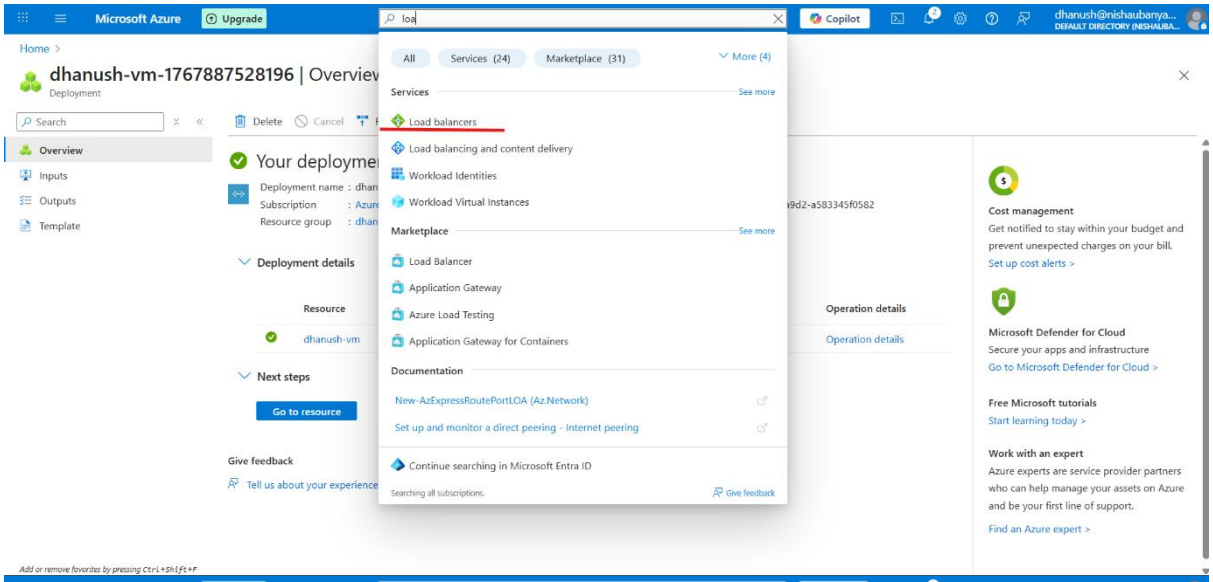

Showing 1 - 3 of 3. Display count: auto
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Microsoft Azure
Upgrade
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Copilot
Home > Network foundation | Virtual networks >
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Virtual network name *
dhanush-vm
Region *
(Europe) Switzerland North
Deploy to an Azure Extended Zone
Previous
Next
Review + create

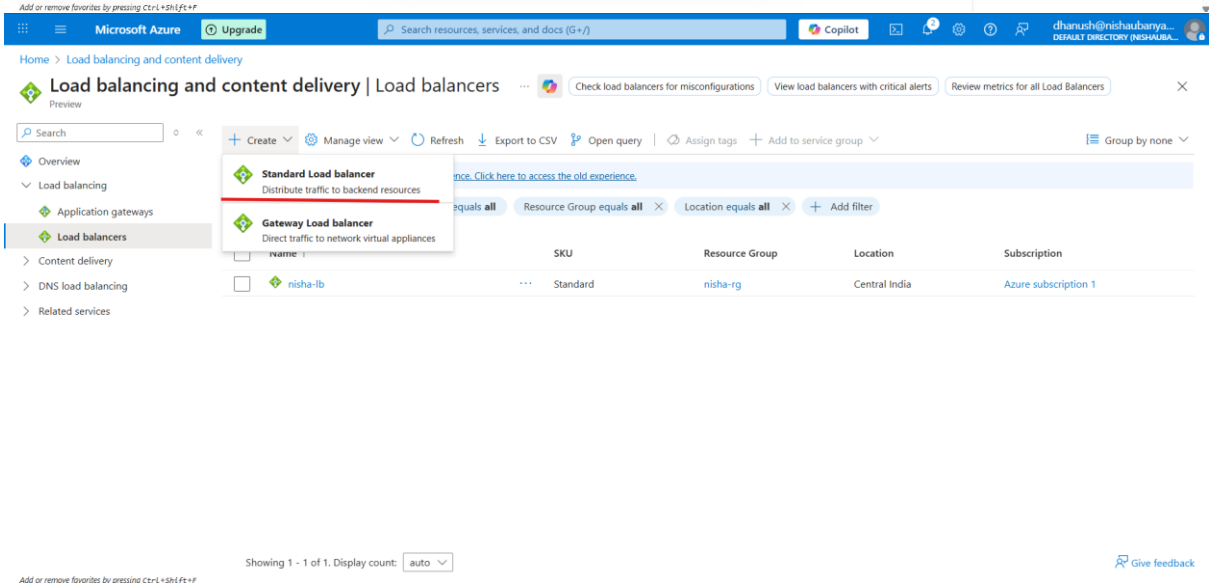

Microsoft Azure
Upgrade
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Copilot
Home > Network foundation | Virtual networks >
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Allocate using IP address pools. Learn more
Add a subnet
10.0.0.0/16
Delete address space
This address prefix overlaps with virtual network 'nisha-vnet'. If you intend to peer these virtual networks, change the address space. Learn more
10.0.0.0
/16
10.0.0.0 - 10.0.255.255
65,536 addresses
Subnets
IP address range
Size
NAT gateway
Add IPv4 address space
Previous
Next
Review + create
Add a subnet
Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more
Subnet purpose
Default
Name *
dhanush-public-subnet
IPv4
Include an IPv4 address space
IPv4 address range
10.0.0.0/16
10.0.0.0 - 10.0.255.255
Starting address *
10.0.1.0
Size
/24 (256 addresses)
Subnet address range
10.0.1.0 - 10.0.1.255
IPv6
Include an IPv6 address space
This virtual network has no IPv6 address ranges.
Private subnet
Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly
Add
Cancel
Give feedback

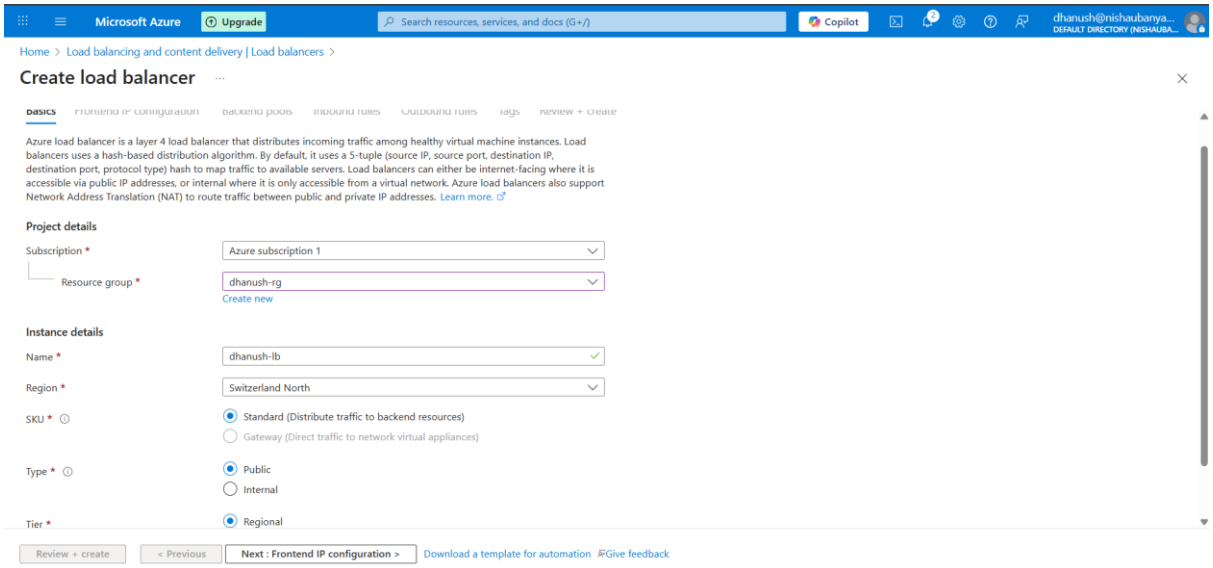


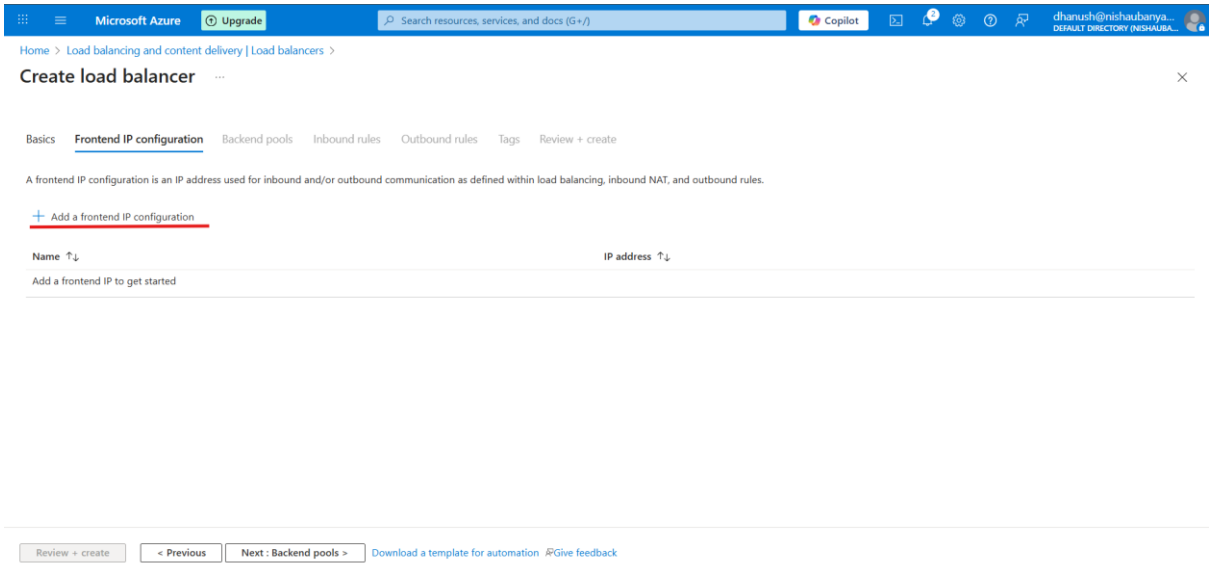


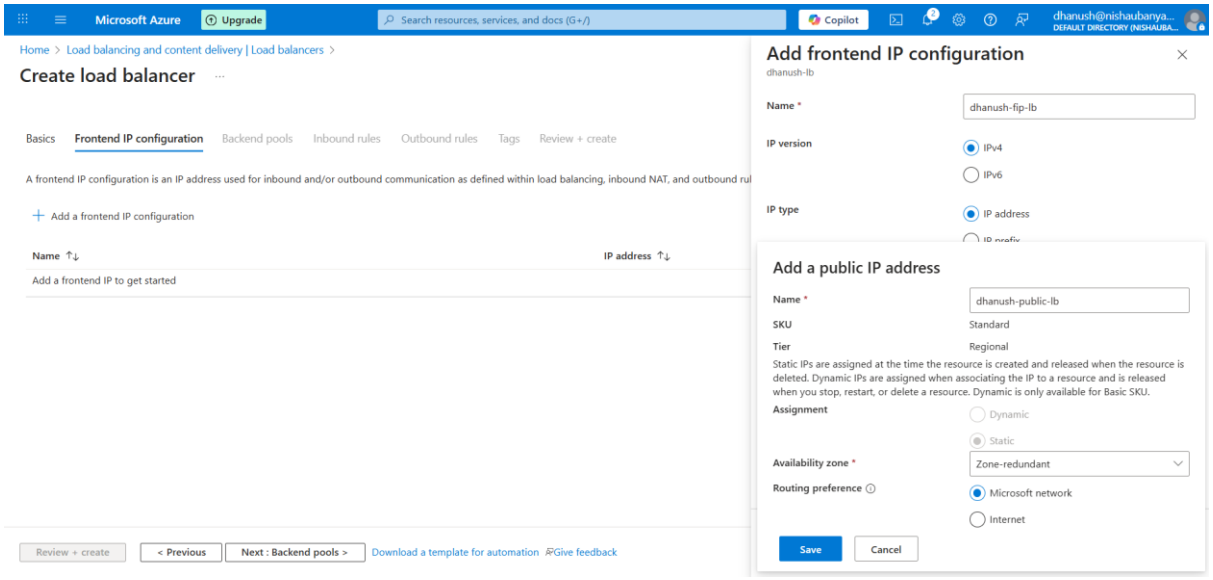
STEP 3 : create a load balancer

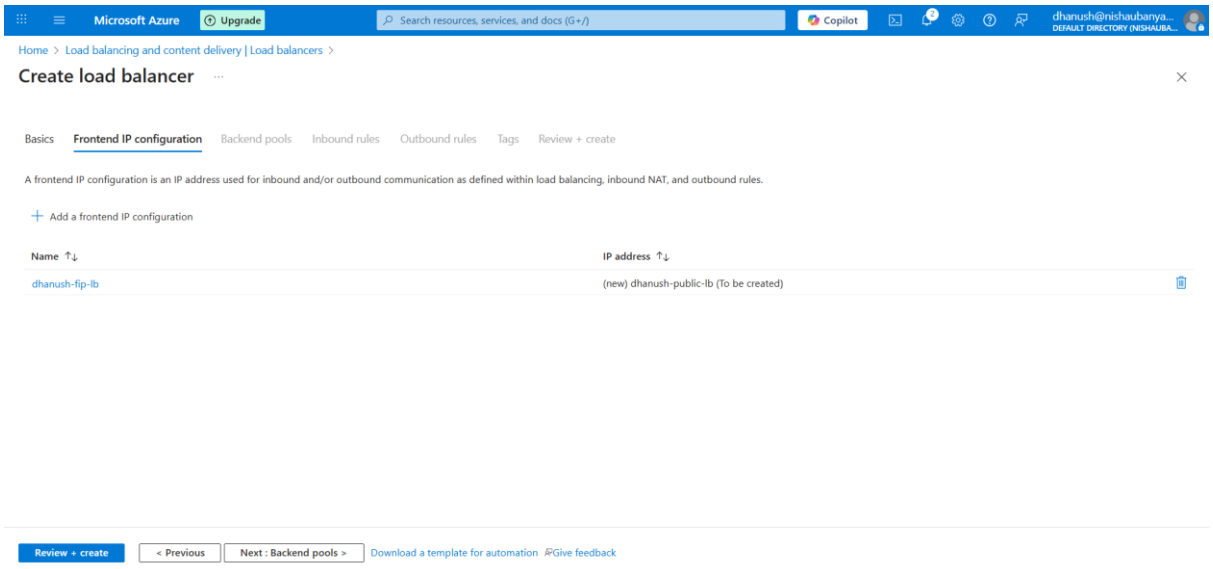

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Home
dhanush-vm-1767887528196 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Your deployment
Deployment name : dhan
Subscription : Azure
Resource group : dhan
Deployment details
Resource
dhanush-vm
Operation details
Operation details
Next steps
Go to resource
Give feedback
Tell us about your experience
All
Services (24)
Marketplace (31)
More (4)
Services
See more
Load balancers
Load balancing and content delivery
Workload Identities
Workload Virtual Instances
Marketplace
See more
Load Balancer
Application Gateway
Azure Load Testing
Application Gateway for Containers
Documentation
New-AzExpressRoutePortLOA (Az.Network)
Set up and monitor a direct peering - Internet peering
Continue searching in Microsoft Entra ID
Searching all subscriptions.
Give feedback
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >
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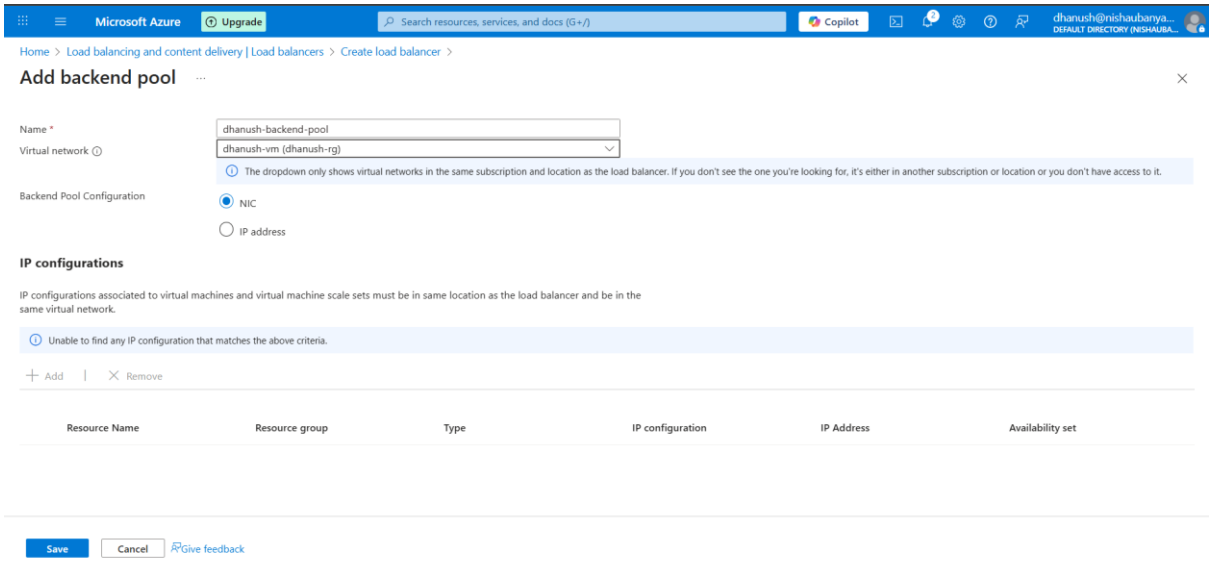

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Load balancing and content delivery | Load balancers
Preview
Check load balancers for misconfigurations
View load balancers with critical alerts
Review metrics for all Load Balancers
Search
Overview
Load balancing
Application gateways
Load balancers
Content delivery
DNS load balancing
Related services
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none
Standard Load balancer
Distribute traffic to backend resources
Gateway Load balancer
Direct traffic to network virtual appliances
nce. Click here to access the old experience.
equals all
Resource Group equals all
Location equals all
Add filter
Name
SKU
Resource Group
Location
Subscription
nisha-lb
Standard
nisha-rg
Central India
Azure subscription 1
Showing 1 - 1 of 1. Display count: auto
Give feedback
Add or remove favorites by pressing Ctrl+Shift+F

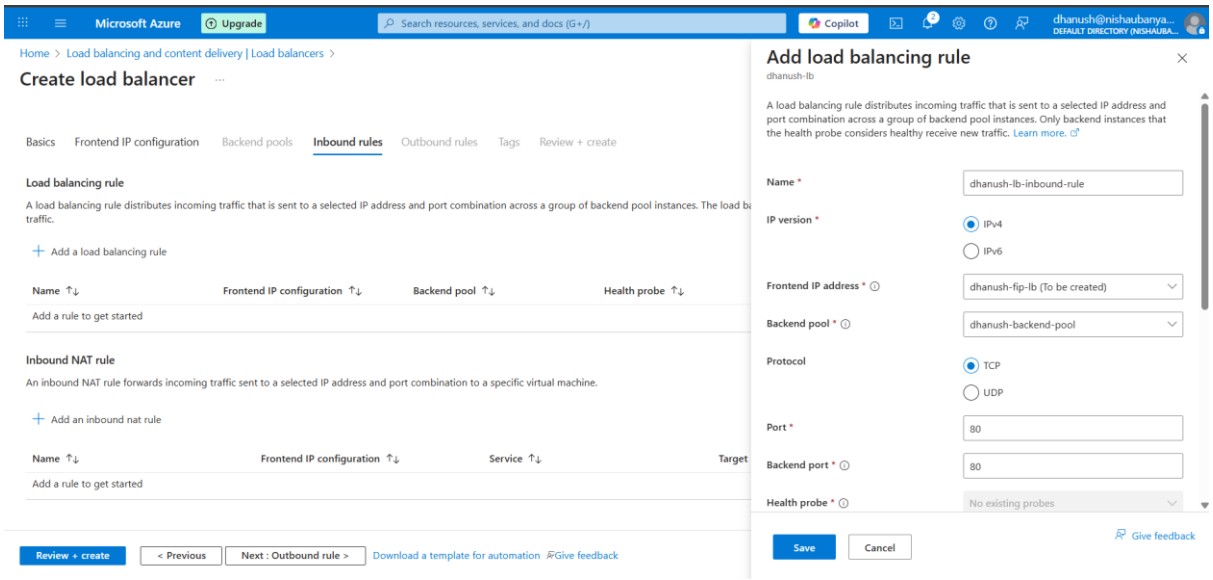

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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Load balancers >
Create load balancer
Basics
Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers uses a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. Learn more.
Project details
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Name *
dhanush-lb
Region *
Switzerland North
SKU *
Standard (Distribute traffic to backend resources)
Gateway (Direct traffic to network virtual appliances)
Type *
Public
Internal
Tier *
Regional
Review + create
< Previous
Next : Frontend IP configuration >
Download a template for automation
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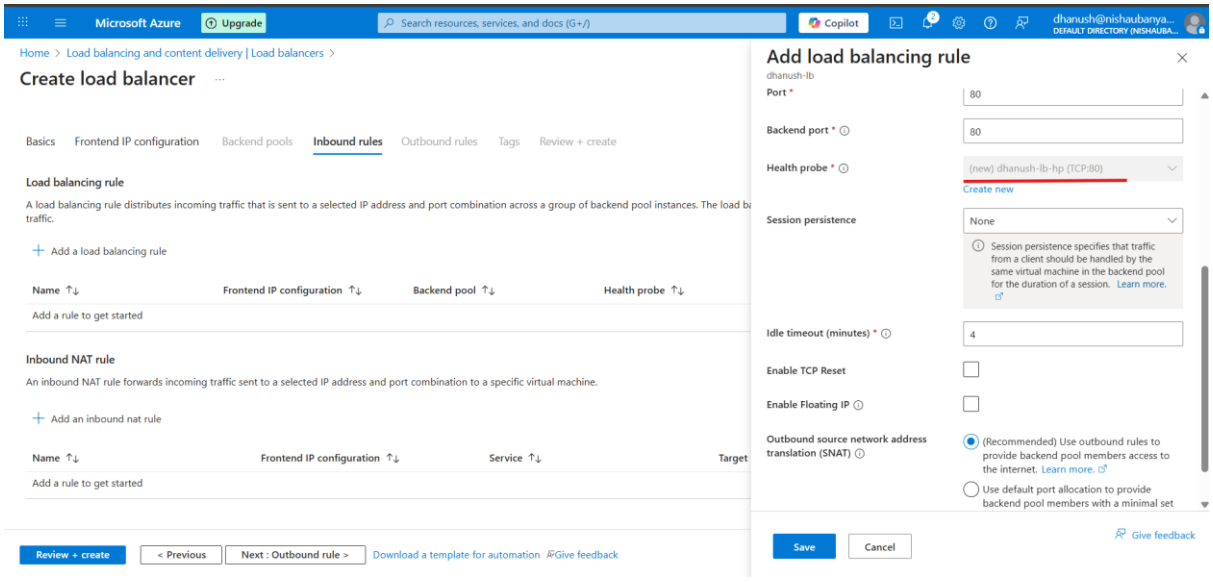

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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Load balancers >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.
Add a frontend IP configuration
Name
IP address
Add a frontend IP to get started
Review + create
< Previous
Next : Backend pools >
Download a template for automation
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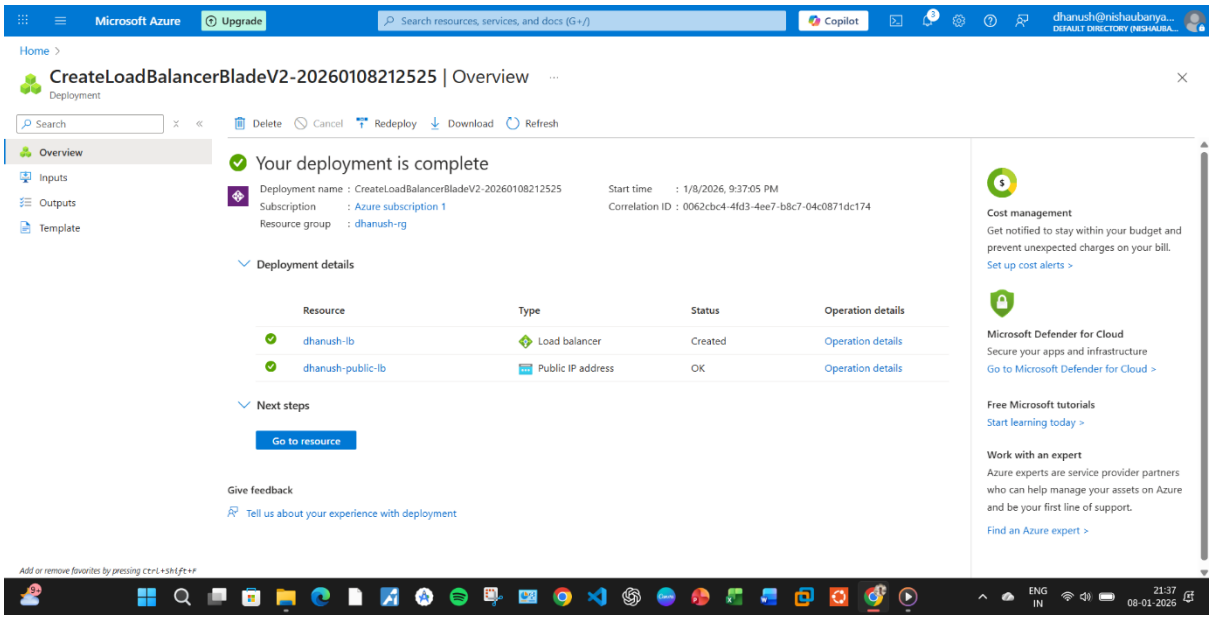

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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Load balancers >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rul
Add a frontend IP configuration
Name
IP address
Add a frontend IP to get started
Add frontend IP configuration
dhanush-lb
Name *
dhanush-fip-lb
IP version
IPv4
IPv6
IP type
IP address
Add a public IP address
Name *
dhanush-public-lb
SKU
Standard
Tier
Regional
Static IPs are assigned at the time the resource is created and released when the resource is deleted. Dynamic IPs are assigned when associating the IP to a resource and is released when you stop, restart, or delete a resource. Dynamic is only available for Basic SKU.
Assignment
Dynamic
Static
Availability zone *
Zone-redundant
Routing preference
Microsoft network
Internet
Save
Cancel
Review + create
< Previous
Next : Backend pools >
Download a template for automation
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dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Load balancers >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.
Add a frontend IP configuration
Name
IP address
dhanush-fip-lb
(new) dhanush-public-lb (To be created)
Review + create
< Previous
Next : Backend pools >
Download a template for automation
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Search resources, services, and docs (G+/)
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dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Load balancers > Create load balancer >
Add backend pool
Name *
dhanush-backend-pool
Virtual network
dhanush-vm (dhanush-rg)
The dropdown only shows virtual networks in the same subscription and location as the load balancer. If you don't see the one you're looking for, it's either in another subscription or location or you don't have access to it.
Backend Pool Configuration
NIC
IP address
IP configurations
IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.
Unable to find any IP configuration that matches the above criteria.
Add
Remove
Resource Name
Resource group
Type
IP configuration
IP Address
Availability set
Save
Cancel
Give feedback


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DEFAULT DIRECTORY (NISHAUBA...
Home > Load balancing and content delivery | Load balancers >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load bi
traffic.
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add load balancing rule
dhanush-lb
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. Only backend instances that the health probe considers healthy receive new traffic. Learn more.
Name *
dhanush-lb-inbound-rule
IP version *
IPv4
IPv6
Frontend IP address *
dhanush-fip-lb (To be created)
Backend pool *
dhanush-backend-pool
Protocol
TCP
UDP
Port *
80
Backend port *
80
Health probe *
No existing probes
Save
Cancel
Give feedback



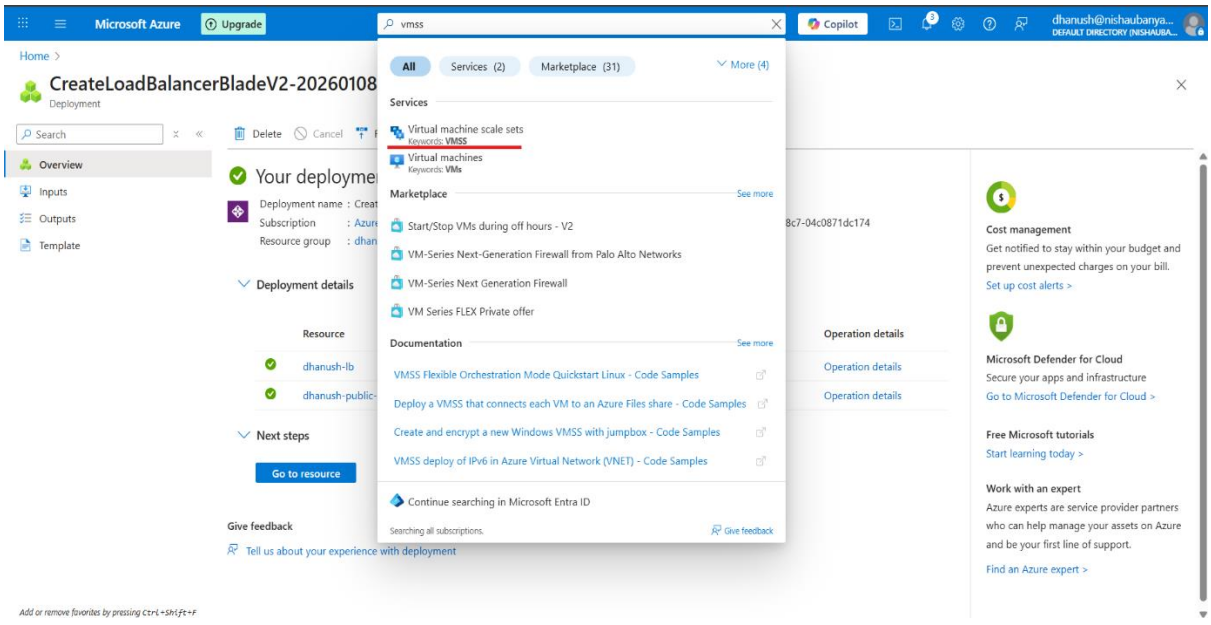


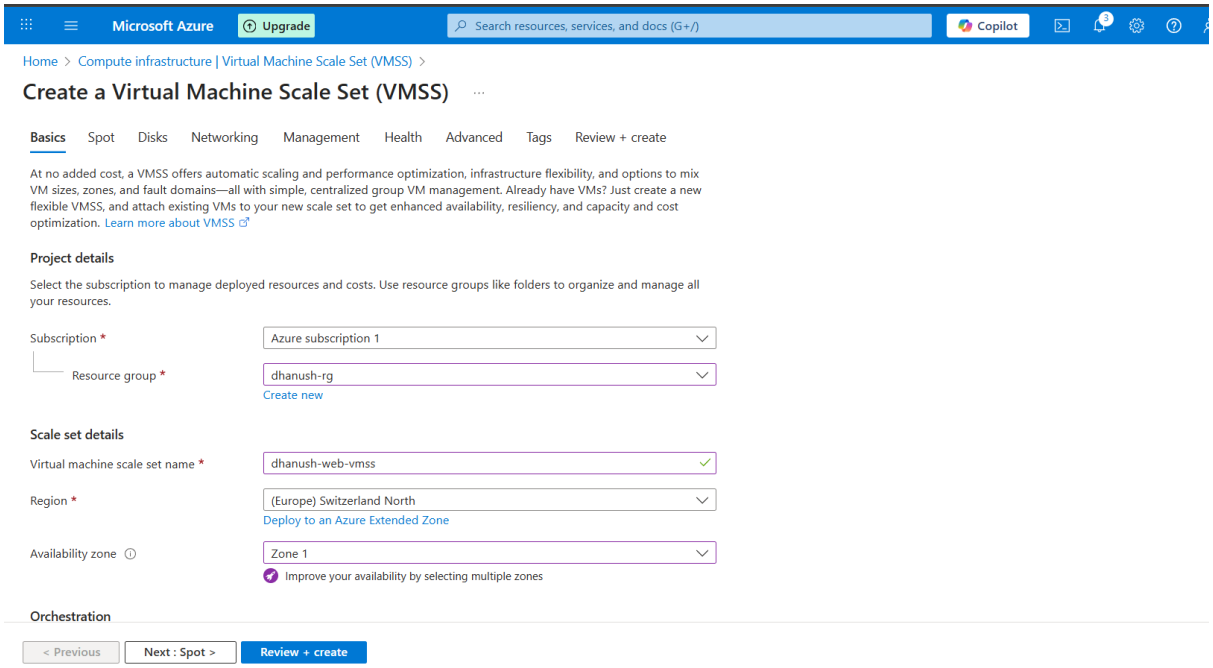
Step 4: Create the VM Scale Set

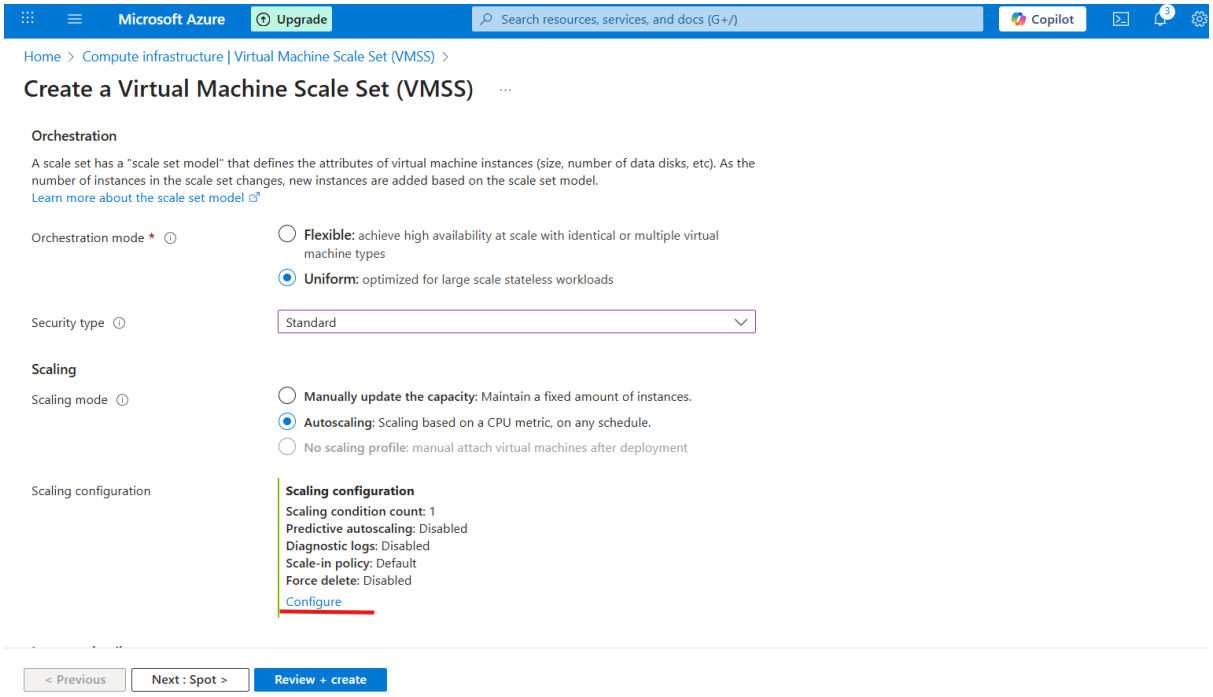
1. In the Azure portal, search **Virtual machine scale sets → + Create**.
2. Basics tab:
   - Subscription: your subscription
   - Resource group: vmss-lab-rg
   - Name: myVMSS

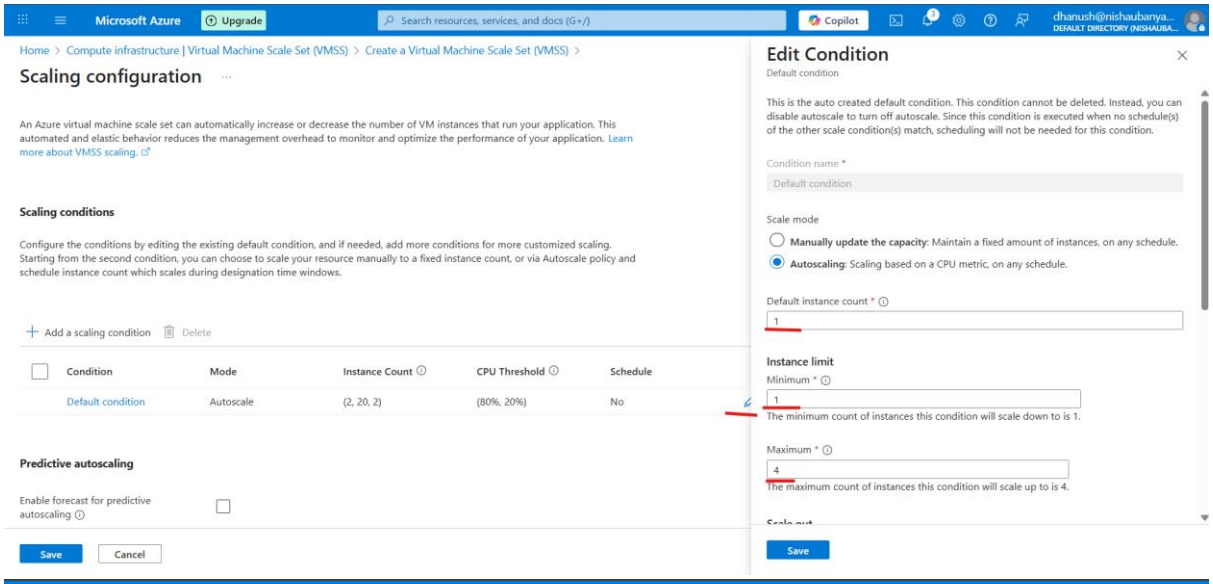
- Region: East US
- Orchestration Mode: **Uniform**
- Image: Ubuntu Server 22.04 LTS (or Windows if you prefer)
- Size: Standard_B1s (small for lab)
- Authentication: Password/SSH

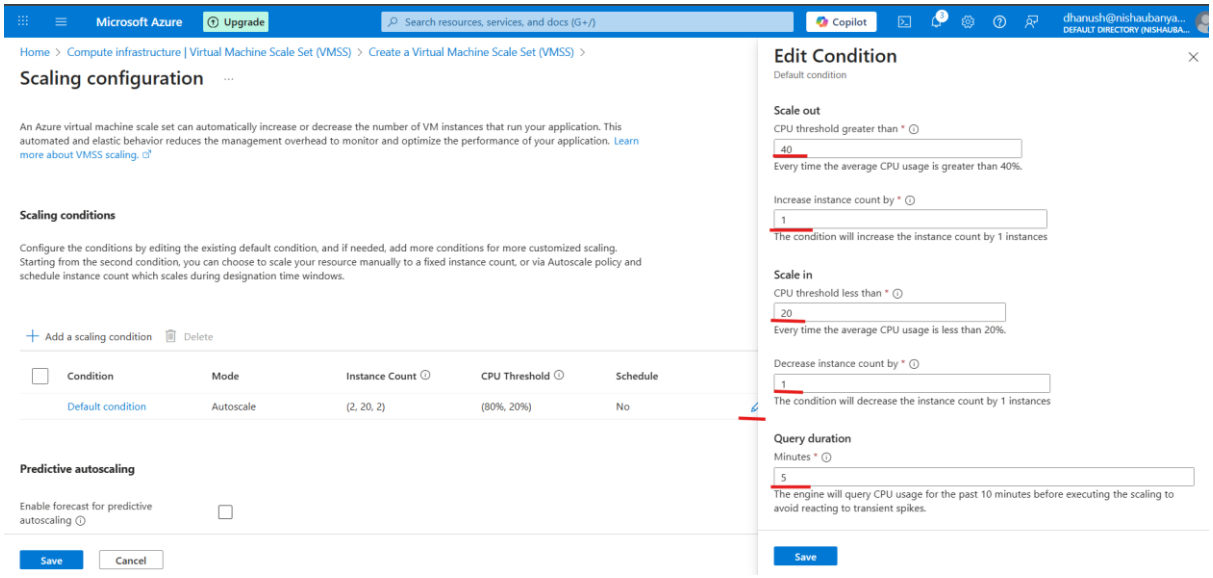
3. Scaling tab:
   - Initial instance count: **2**
   - Scaling: Enable autoscale

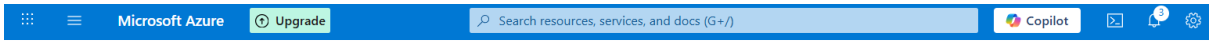



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Copilot
Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) >
Create a Virtual Machine Scale Set (VMSS)
Basics Spot Disks Networking Management Health Advanced Tags Review + create
At no added cost, a VMSS offers automatic scaling and performance optimization, infrastructure flexibility, and options to mix VM sizes, zones, and fault domains—all with simple, centralized group VM management. Already have VMs? Just create a new flexible VMSS, and attach existing VMs to your new scale set to get enhanced availability, resiliency, and capacity and cost optimization. Learn more about VMSS
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription * Azure subscription 1
Resource group * dhanush-rg
Create new
Scale set details
Virtual machine scale set name * dhanush-web-vmss
Region * (Europe) Switzerland North
Deploy to an Azure Extended Zone
Availability zone Zone 1
Improve your availability by selecting multiple zones
Orchestration
< Previous Next : Spot > Review + create


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Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) >
Create a Virtual Machine Scale Set (VMSS)
Orchestration
A scale set has a "scale set model" that defines the attributes of virtual machine instances (size, number of data disks, etc). As the number of instances in the scale set changes, new instances are added based on the scale set model.
Learn more about the scale set model
Orchestration mode * Flexible: achieve high availability at scale with identical or multiple virtual machine types
Uniform: optimized for large scale stateless workloads
Security type Standard
Scaling
Scaling mode Manually update the capacity: Maintain a fixed amount of instances.
Autoscaling: Scaling based on a CPU metric, on any schedule.
No scaling profile: manual attach virtual machines after deployment
Scaling configuration
Scaling configuration
Scaling condition count: 1
Predictive autoscaling: Disabled
Diagnostic logs: Disabled
Scale-in policy: Default
Force delete: Disabled
Configure
< Previous Next : Spot > Review + create


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DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) > Create a Virtual Machine Scale Set (VMSS) >
Scaling configuration
An Azure virtual machine scale set can automatically increase or decrease the number of VM instances that run your application. This automated and elastic behavior reduces the management overhead to monitor and optimize the performance of your application. Learn more about VMSS scaling.
Scaling conditions
Configure the conditions by editing the existing default condition, and if needed, add more conditions for more customized scaling. Starting from the second condition, you can choose to scale your resource manually to a fixed instance count, or via Autoscale policy and schedule instance count which scales during designation time windows.
Add a scaling condition
Delete
Condition
Mode
Instance Count
CPU Threshold
Schedule
Default condition
Autoscale
(2, 20, 2)
(80%, 20%)
No
Predictive autoscaling
Enable forecast for predictive autoscaling
Save
Cancel
Edit Condition
Default condition
This is the auto created default condition. This condition cannot be deleted. Instead, you can disable autoscale to turn off autoscale. Since this condition is executed when no schedule(s) of the other scale condition(s) match, scheduling will not be needed for this condition.
Condition name *
Default condition
Scale mode
Manually update the capacity: Maintain a fixed amount of instances, on any schedule.
Autoscaling: Scaling based on a CPU metric, on any schedule.
Default instance count *
1
Instance limit
Minimum *
1
The minimum count of instances this condition will scale down to is 1.
Maximum *
4
The maximum count of instances this condition will scale up to is 4.
Save


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Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) > Create a Virtual Machine Scale Set (VMSS) >
Scaling configuration
An Azure virtual machine scale set can automatically increase or decrease the number of VM instances that run your application. This automated and elastic behavior reduces the management overhead to monitor and optimize the performance of your application. Learn more about VMSS scaling.
Scaling conditions
Configure the conditions by editing the existing default condition, and if needed, add more conditions for more customized scaling. Starting from the second condition, you can choose to scale your resource manually to a fixed instance count, or via Autoscale policy and schedule instance count which scales during designation time windows.
Add a scaling condition
Delete
Condition
Mode
Instance Count
CPU Threshold
Schedule
Default condition
Autoscale
(2, 20, 2)
(80%, 20%)
No
Predictive autoscaling
Enable forecast for predictive autoscaling
Save
Cancel
Edit Condition
Default condition
Scale out
CPU threshold greater than *
40
Every time the average CPU usage is greater than 40%.
Increase instance count by *
1
The condition will increase the instance count by 1 instances
Scale in
CPU threshold less than *
20
Every time the average CPU usage is less than 20%.
Decrease instance count by *
1
The condition will decrease the instance count by 1 instances
Query duration
Minutes *
5
The engine will query CPU usage for the past 10 minutes before executing the scaling to avoid reacting to transient spikes.
Save



Scaling configuration

An Azure virtual machine scale set can automatically increase or decrease the number of VM instances that run your application. This automated and elastic behavior reduces the management overhead to monitor and optimize the performance of your application. Learn more about VMSS scaling.

Scaling conditions

Configure the conditions by editing the existing default condition, and if needed, add more conditions for more customized scaling. Starting from the second condition, you can choose to scale your resource manually to a fixed instance count, or via Autoscale policy and schedule instance count which scales during designation time windows.

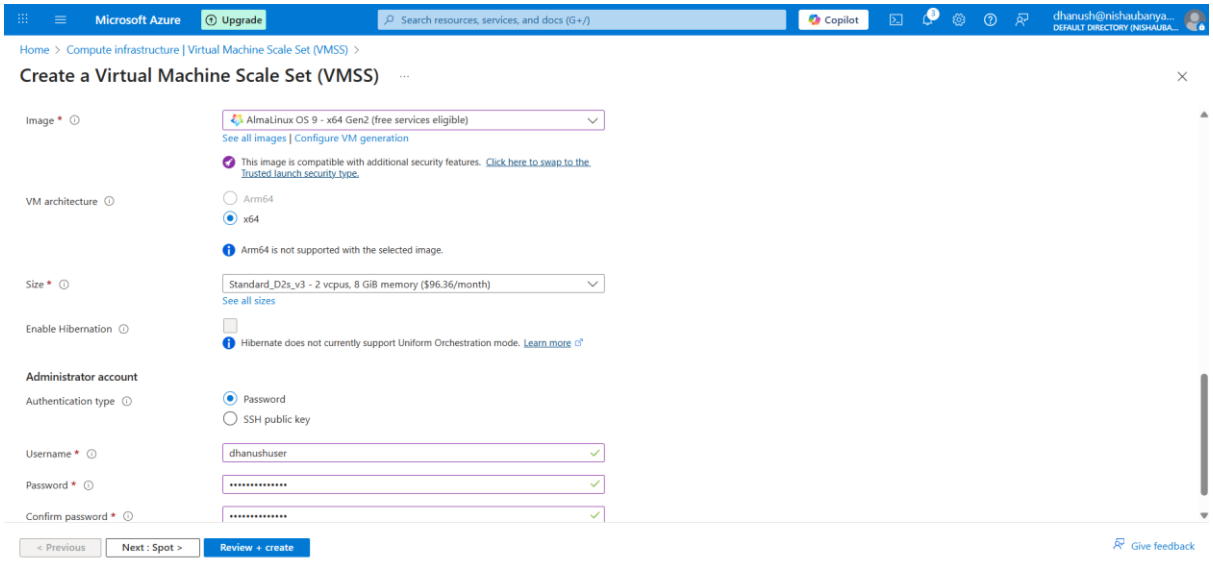
+ Add a scaling condition Delete

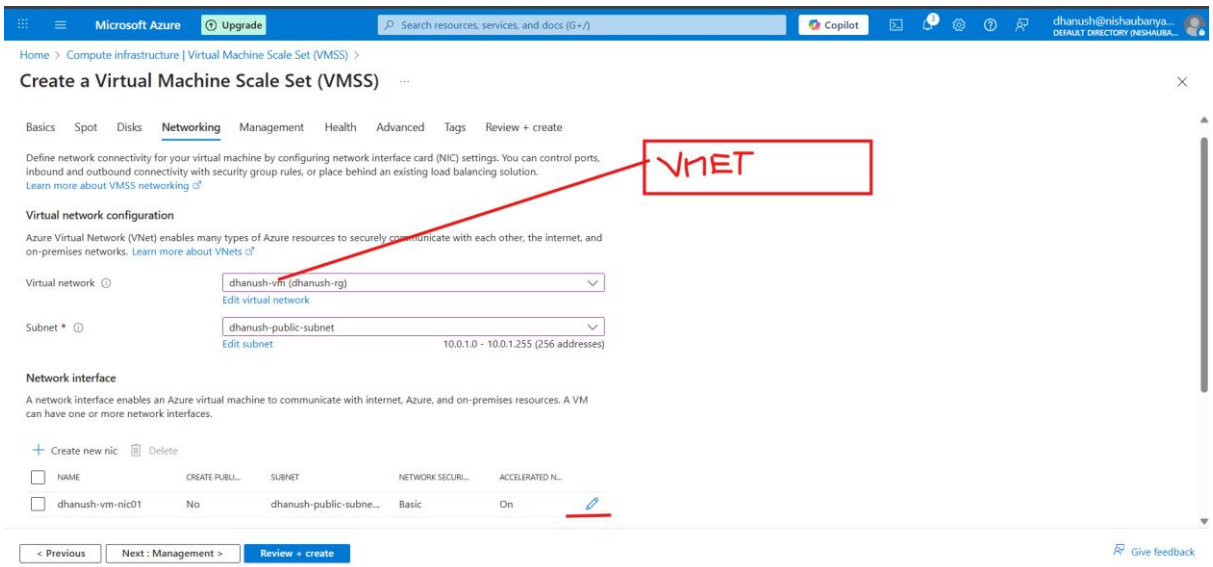
Condition	Mode	Instance Count	CPU Threshold	Schedule
Default condition	Autoscale	(1, 4, 1)	(40%, 20%)	No

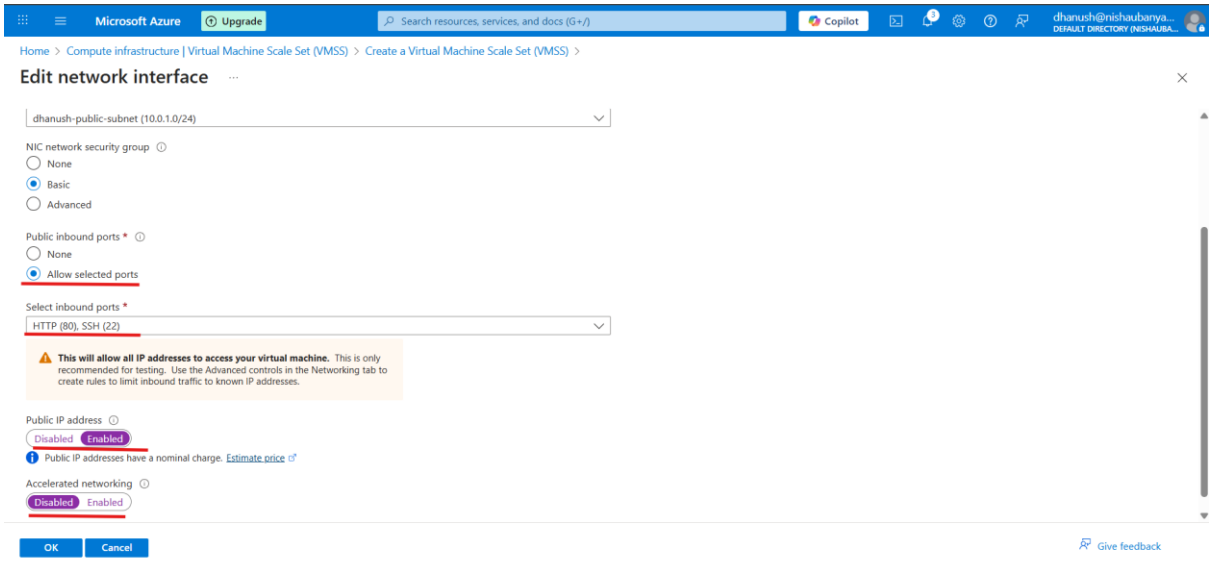
Predictive autoscaling

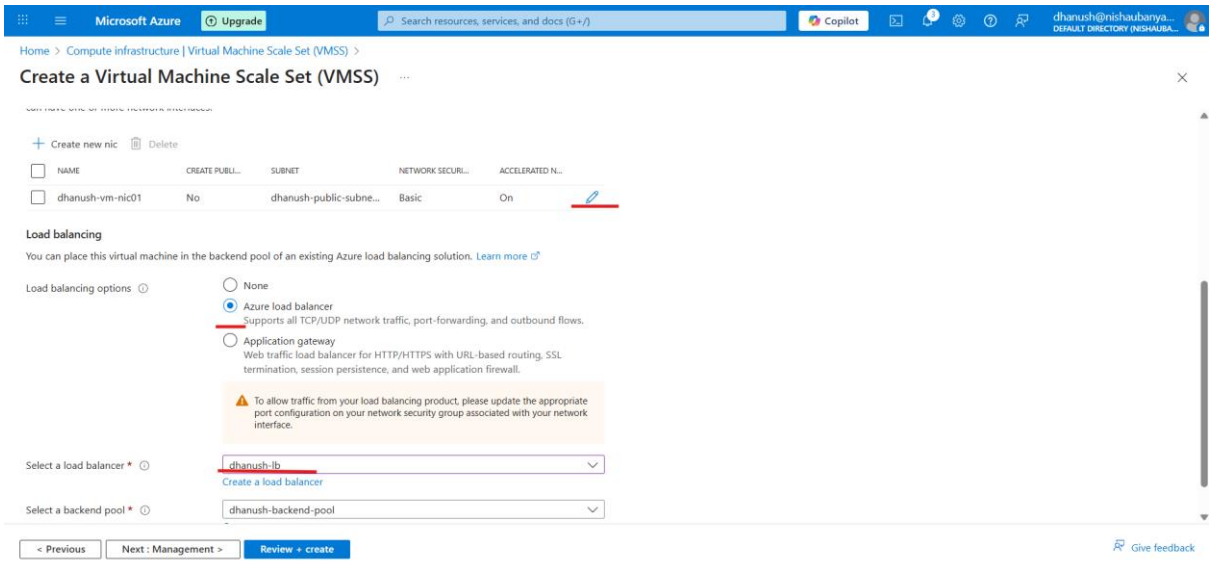
Enable forecast for predictive autoscaling

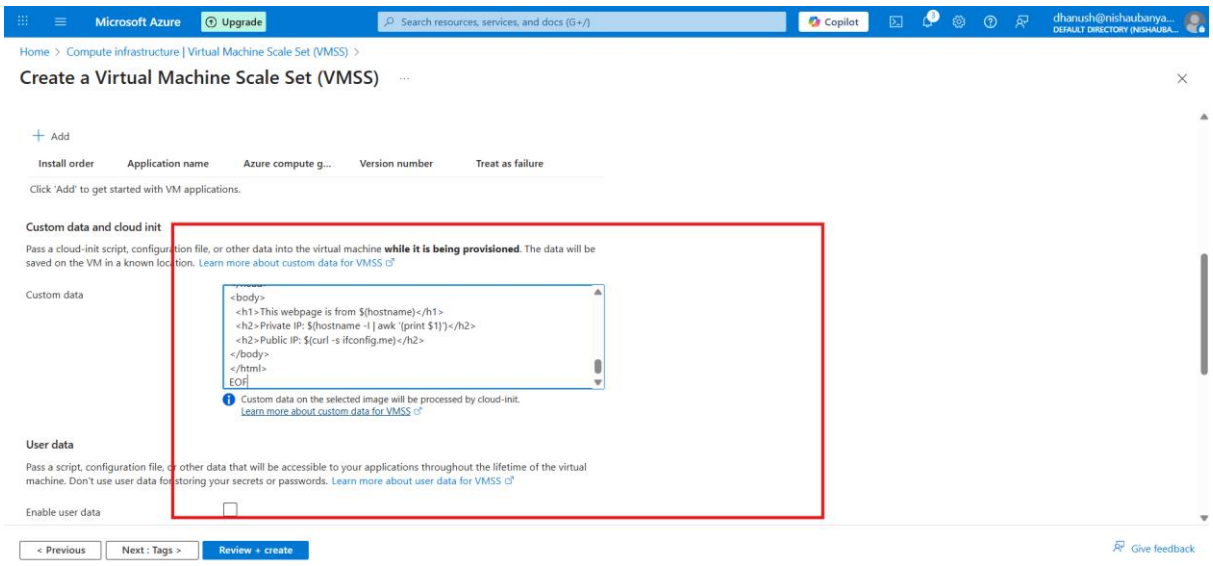
Save Cancel


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) >
Create a Virtual Machine Scale Set (VMSS)
Image *
AlmaLinux OS 9 - x64 Gen2 (free services eligible)
See all images | Configure VM generation
This image is compatible with additional security features. Click here to swap to the Trusted launch security type.
VM architecture
Arm64
x64
Arm64 is not supported with the selected image.
Size *
Standard_D2s_v3 - 2 vcpus, 8 GiB memory ($96.36/month)
See all sizes
Enable Hibernation
Hibernate does not currently support Uniform Orchestration mode. Learn more
Administrator account
Authentication type
Password
SSH public key
Username *
dhanushuser
Password *
Confirm password *
< Previous
Next : Spot >
Review + create
Give feedback


Microsoft Azure
Upgrade
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Copilot
dhanush@nishaubanya...
DEFAULT DIRECTORY (NISHAUBA...
Home > Compute infrastructure | Virtual Machine Scale Set (VMSS) >
Create a Virtual Machine Scale Set (VMSS)
Basics
Spot
Disks
Networking
Management
Health
Advanced
Tags
Review + create
Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more about VMSS networking
VNET
Virtual network configuration
Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. Learn more about VNets
Virtual network
dhanush-vm (dhanush-rg)
Edit virtual network
Subnet *
dhanush-public-subnet
Edit subnet
10.0.1.0 - 10.0.1.255 (256 addresses)
Network interface
A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.
Create new nic
Delete
NAME
CREATE PUBLI...
SUBNET
NETWORK SECURI...
ACCELERATED N...
dhanush-vm-nic01
No
dhanush-public-subne...
Basic
On
< Previous
Next : Management >
Review + create
Give feedback







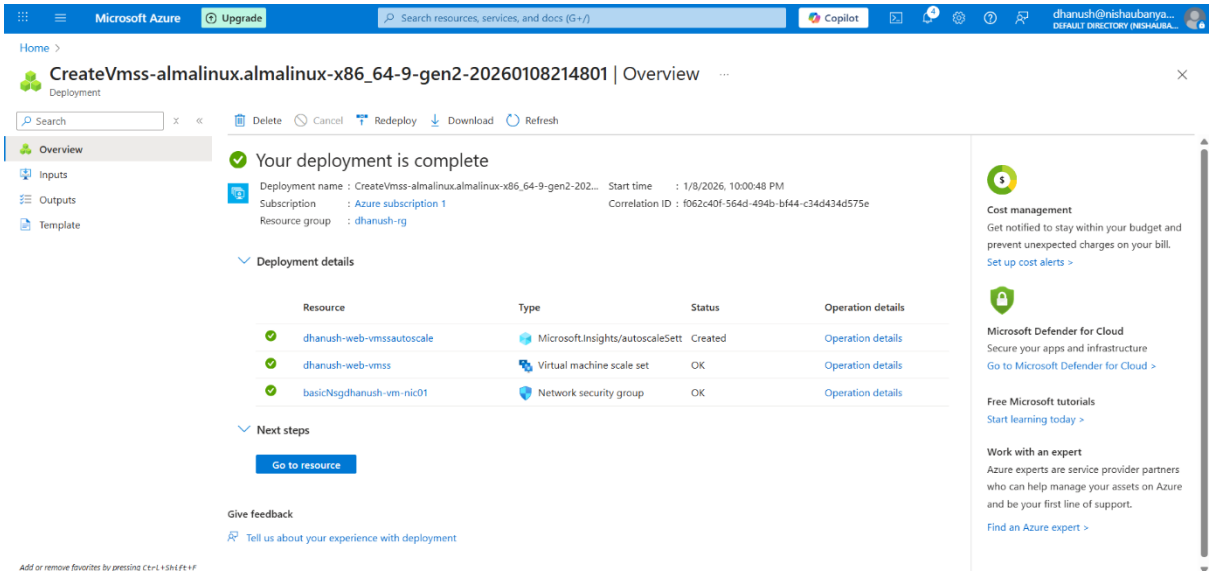
Custom data

```
#!/bin/bash
```

```
# Install Apache HTTP server
dnf install -y httpd curl

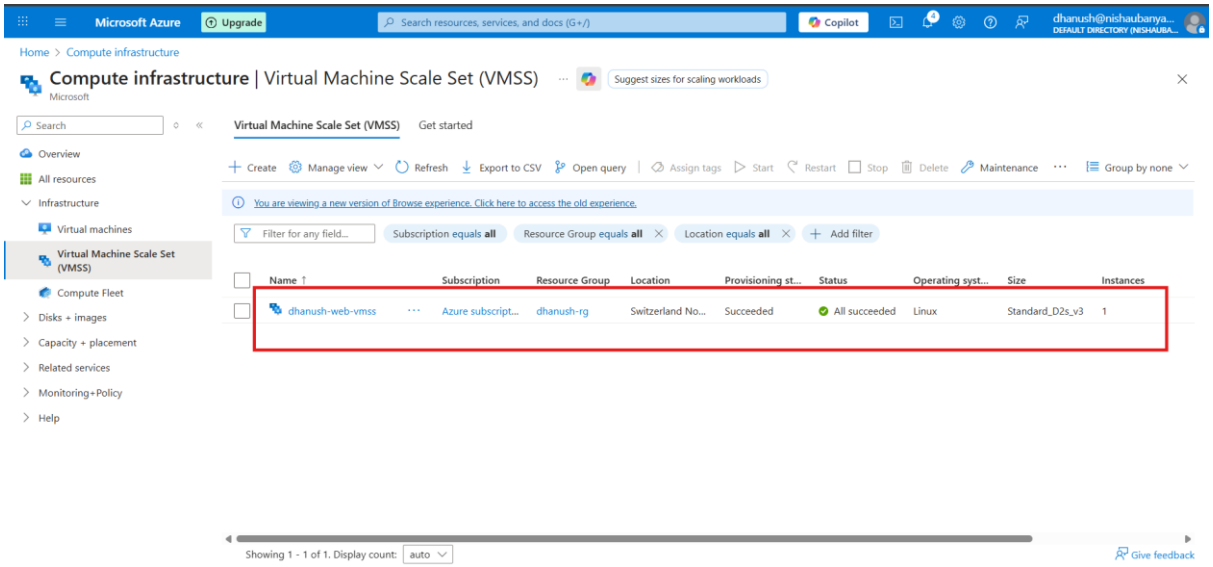
# Enable and start Apache
systemctl enable httpd
systemctl start httpd

# Create index.html
cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Azure VM Webpage</title>
  <style>
    body {
      background-color: green;
      color: white;
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 10%;
    }
    h1 {
      font-size: 48px;
    }
    h2 {
      font-size: 28px;
    }
  </style>
</head>
<body>
  <h1>This webpage is from $(hostname)</h1>
  <h2>Private IP: $(hostname -I | awk '{print $1}')</h2>
  <h2>Public IP: $(curl -s ifconfig.me)</h2>
</body>
</html>
EOF
```



Copy and paste public ip to the browser





Go to the cmd and connect the machine

Increasing a load on virtual machin

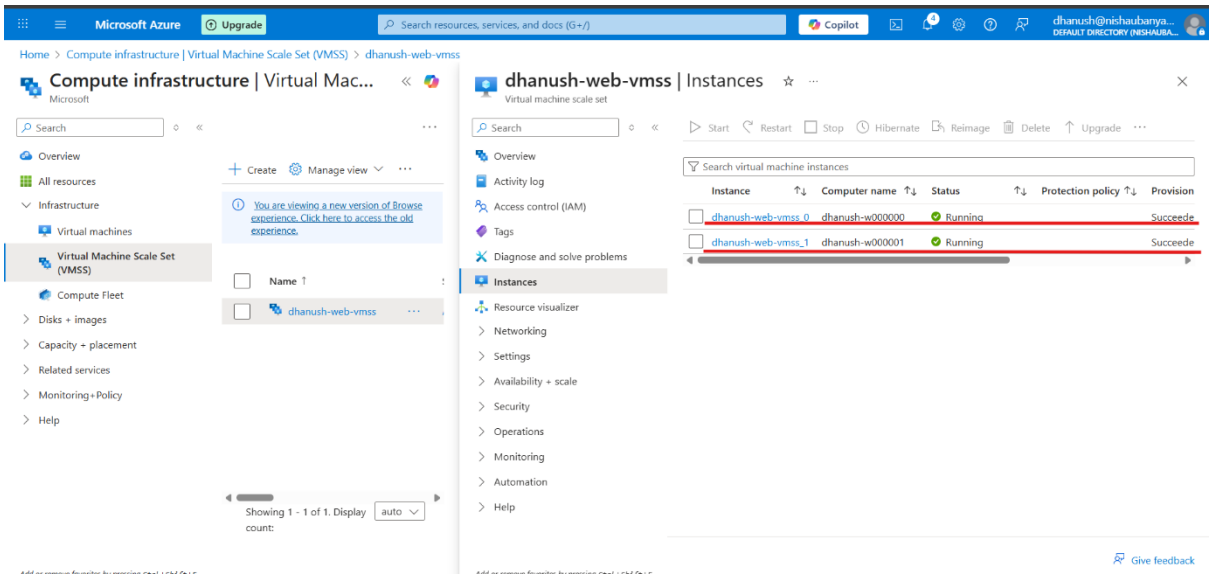
To using command (yes > /dev/null &)

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DHANUSH>ssh dhanushuser@20.199.169.53
The authenticity of host '20.199.169.53 (20.199.169.53)' can't be established.
ED25519 key fingerprint is SHA256:4HleFkYZ23x0mnaxJ4se9bAIXILuaRczWHjvwPkamRA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.199.169.53' (ED25519) to the list of known hosts.
dhanushuser@20.199.169.53's password:
[dhanushuser@dhanush-w000000 ~]$ sudo -i
[root@dhanush-w000000 ~]# yes > /dev/null &
[1] 2376
[root@dhanush-w000000 ~]# yes > /dev/null &
[2] 2377
[root@dhanush-w000000 ~]# yes > /dev/null &
[3] 2378
[root@dhanush-w000000 ~]# yes > /dev/null &
[4] 2379
[root@dhanush-w000000 ~]# yes > /dev/null &
[5] 2380
[root@dhanush-w000000 ~]# yes > /dev/null &
[6] 2381
[root@dhanush-w000000 ~]#
[root@dhanush-w000000 ~]# yes > /dev/null &
[7] 2382
[root@dhanush-w000000 ~]# yes > /dev/null &
[8] 2383
[root@dhanush-w000000 ~]# yes > /dev/null &
[9] 2384
[root@dhanush-w000000 ~]# yes > /dev/null &
[10] 2385
[root@dhanush-w000000 ~]# yes > /dev/null &
[11] 2386
[root@dhanush-w000000 ~]# yes > /dev/null &
[12] 2387
[root@dhanush-w000000 ~]# yes > /dev/null &
[13] 2392
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[14] 2403
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
```

```
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[14] 2403
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[15] 2404
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[16] 2405
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[17] 2406
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[18] 2407
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[19] 2408
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[20] 2413
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[21] 2414
[root@dhanush-w000000 ~]# dd if=/dev/zero of=/dev/null &
[22] 2415
[root@dhanush-w000000 ~]# w
 16:49:19 up 17 min,  1 user,  load average: 13.13, 4.45, 1.59
USER     TTY        LOGIN@   IDLE   JCPU   PCPU WHAT
dhanushu pts/0     16:45    6.00s  3:52   0.04s sshd: dhanushuser [priv]
[root@dhanush-w000000 ~]# top
top - 16:49:42 up 18 min,  1 user,  load average: 16.16, 5.86, 2.14
Tasks: 158 total,  23 running, 135 sleeping,   0 stopped,   0 zombie
%Cpu(s): 54.1 us, 45.9 sy,  0.0 ni,  0.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :   7936.7 total,   7412.2 free,    520.7 used,    266.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   7416.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+
   2377 root      20   0    5592   1912   1780 R  10.5   0.0   0:24.63
   2382 root      20   0    5592   1912   1784 R  10.5   0.0   0:17.18
   2383 root      20   0    5592   1912   1784 R  10.5   0.0   0:16.70
   2387 root      20   0    5592   1908   1780 R  10.5   0.0   0:15.81
   2392 root      20   0    5592   1912   1784 R  10.5   0.0   0:15.58
   2403 root      20   0    5632   1960   1824 R  10.5   0.0   0:05.32
   2404 root      20   0    5632   1944   1812 R  10.5   0.0   0:05.04
   2406 root      20   0    5632   1956   1820 R  10.5   0.0   0:04.55
   2407 root      20   0    5632   1952   1820 R  10.5   0.0   0:04.41
   2413 root      20   0    5632   1944   1812 R  10.5   0.0   0:02.57
   2415 root      20   0    5632   1956   1824 R  10.5   0.0   0:02.37
```



To decreasing a load and removing a automatically delete vm

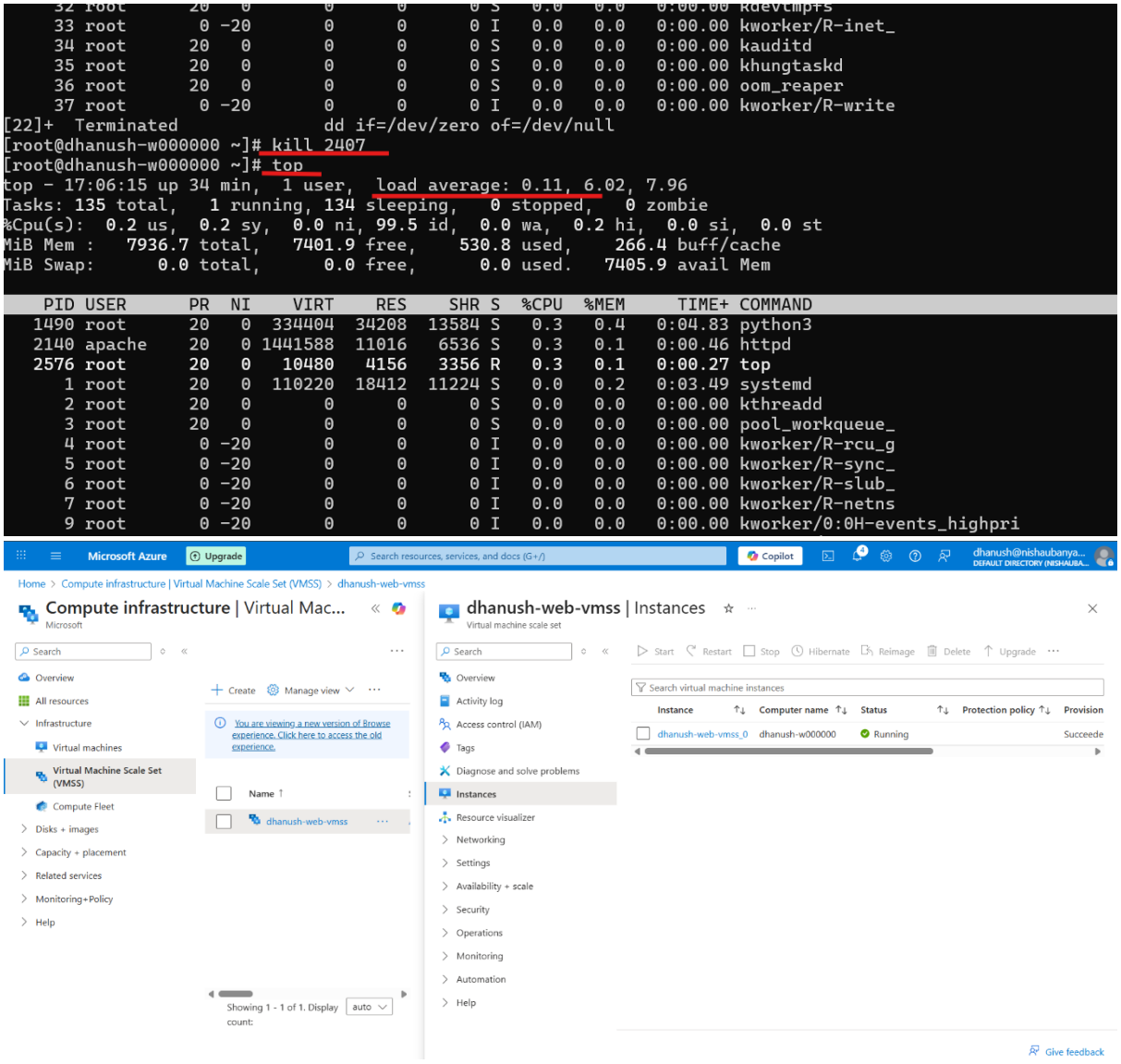
To using a (w) comment and checking

```
[root@dhanush-w000000 ~]# kill 2387
[root@dhanush-w000000 ~]# kill 2378
[12]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2379
[3]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2380
[4]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2382
[5]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 238
-bash: kill: (238) - No such process
[7]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# top
top - 16:59:09 up 27 min,  1 user,  load average: 18.38, 18.49, 10.84
Tasks: 149 total,  17 running, 132 sleeping,   0 stopped,   0 zombie
%Cpu(s): 56.5 us, 42.6 sy,  0.0 ni,  0.0 id,  0.0 wa,  0.7 hi,  0.2 si,  0.0 st
MiB Mem :   7936.7 total,   7405.3 free,    527.6 used,    266.2 buff/cache
```

```
[root@dhanush-w000000 ~]# kill 2381
[root@dhanush-w000000 ~]# kill 2383
[6]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2384
[8]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2385
[9]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2405
[10]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2408
[root@dhanush-w000000 ~]# kill 2408
-bash: kill: (2408) - No such process
[16]   Terminated              dd if=/dev/zero of=/dev/null
[19]   Terminated              dd if=/dev/zero of=/dev/null
[root@dhanush-w000000 ~]# kill 2414
[root@dhanush-w000000 ~]# kill 2406
[21]-  Terminated              dd if=/dev/zero of=/dev/null
[root@dhanush-w000000 ~]# top
top - 17:00:46 up 29 min,  1 user,  load average: 11.93, 16.48, 10.88
Tasks: 142 total,   9 running, 133 sleeping,   0 stopped,   0 zombie
%Cpu(s): 57.6 us, 41.9 sy,  0.0 ni,  0.0 id,  0.0 wa,  0.5 hi,  0.0 si,  0.0 st
MiB Mem :   7936.7 total,   7403.9 free,    529.0 used,    266.2 buff/cache
```

```
    28 root      20   0       0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/1
[17]   Terminated              dd if=/dev/zero of=/dev/null
[root@dhanush-w000000 ~]# kill 2376
[root@dhanush-w000000 ~]# kill 2386
[1]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2392
[11]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# 2403
-bash: 2403: command not found
[13]   Terminated              yes > /dev/null
[root@dhanush-w000000 ~]# kill 2404
[root@dhanush-w000000 ~]# kill 2403
[15]   Terminated              dd if=/dev/zero of=/dev/null
[root@dhanush-w000000 ~]# kill 2413
[14]   Terminated              dd if=/dev/zero of=/dev/null
[root@dhanush-w000000 ~]# kill 2415
[20]-  Terminated              dd if=/dev/zero of=/dev/null
[root@dhanush-w000000 ~]# top
top - 17:02:42 up 31 min,  1 user,  load average: 3.69, 12.14, 9.99
Tasks: 136 total,   2 running, 134 sleeping,   0 stopped,   0 zombie
%Cpu(s): 31.1 us, 19.1 sy,  0.0 ni, 49.5 id,  0.0 wa,  0.3 hi,  0.0 si,  0.0 st
MiB Mem :   7936.7 total,   7402.0 free,    530.7 used,    266.4 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   7406.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2407 root      20   0    5632   1952   1820 R  99.3   0.0   2:30.85 dd
   2518 root      20   0       0      0      0 I   0.3   0.0   0:00.16 kworker/1:2-events
      1 root      20   0  110220  18412  11224 S   0.0   0.2   0:03.48 systemd
```



✅ Prerequisites

- Azure subscription
- Azure Portal access (we'll use **GUI**)
- Alternatively, you can run via **Azure CLI**

Step 1: Create a Resource Group

Azure Portal

1. Go to **Resource groups → + Create**.
2. Name: vmss-lab-rg
3. Region: Choose nearest region (e.g., East US).
4. Review + Create.

Step 2: Create the VM Scale Set

1. In the Azure portal, search **Virtual machine scale sets → + Create**.
2. Basics tab:
    - Subscription: your subscription
    - Resource group: vmss-lab-rg
    - Name: myVMSS
    - Region: East US
    - Orchestration Mode: **Uniform**
    - Image: Ubuntu Server 22.04 LTS (or Windows if you prefer)
    - Size: Standard_B1s (small for lab)
    - Authentication: Password/SSH

3. Scaling tab:
    - Initial instance count: **2**
    - Scaling: Enable autoscale

Step 3: Configure Autoscale Rules

1. Go to **Scaling → Custom autoscale**.
2. Add a **rule for CPU**:
    - Metric: Percentage CPU
    - Condition: Greater than 70%
    - Action: Increase by 1 instance
    - Cooldown: 5 minutes
3. Add a **rule for Memory** (Memory metric needs **Azure Monitor Diagnostic extension** → you must enable it):
    - Metric namespace: Guest (Classic)
    - Metric: Available Memory
    - Condition: Less than 500 MB
    - Action: Increase by 1 instance
    - Cooldown: 5 minutes

4. Add a **scale in rule**:
    - Metric: Percentage CPU
    - Condition: Less than 30%
    - Action: Decrease by 1 instance
    - Cooldown: 10 minutes
5. **Limits**:
    - Minimum: 2
    - Maximum: 5

Step 4: Networking

- Select **Create new Load Balancer** (for inbound traffic).
- Leave defaults for simplicity.

Step 5: Review + Create

- Validate and click **Create**.
- Wait for deployment (2–3 min).

Step 6: Generate CPU/Memory Load (to trigger scale out)

1. Connect to one of the VMs in the VMSS via SSH.
2. Install **stress tool**:
3. sudo apt update

sudo apt install stress -y

4. Run stress test (CPU load):

stress --cpu 2 --timeout 300

5. Or memory stress:

stress --vm 1 --vm-bytes 600M --timeout 300

6. Monitor VMSS → **Instances tab**. You'll see it scale from 2 → 3 → … → max 5 VMs.

Step 7: Verify Autoscaling

- After load stops, CPU/Memory drops → VMSS will automatically **scale in** back toward 2 VMs.
- You can confirm in **Metrics (Azure Monitor)**.

Cleanup

To avoid charges:

delete RG

✅ **End Result:** You created a **VM Scale Set** with 2 initial VMs, autoscaling up to 5 based on **CPU and Memory** load.